TITLE PAGE

- **Problem Statement ID – 26261**
- **Problem Statement Title - AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification**
- **Theme - Smart Automation**
- **PS Category- Software**
- **Team ID- 198045**
- **Team Name – MARK 44**

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ShowYourWork

The AI Document Checker That Never Accuses, Only Shows Evidence

Flag. Verify. Decide.

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Problem in the statement	Our answer
Missing, expired, inconsistent documents	Graph and date checks catch them early
Repeated manual checking	Officers see only exceptions
Suspected alterations	Evidence-linked “indicator” flags
Issuer confirmation	Verification Ladder with fallbacks
Infrastructure gaps	Flag “needs evidence” or “needs inspection”
Trust and traceability	Evidence Cards, audit trail, officer sign-off

How it works

Extract: OCR + layout parsing turns each document into **Evidence Nodes** (value, page, bounding box, confidence score)

Cross-check: nodes are linked in an **Evidence Graph** across documents and AICTE records

Validate: a **JSON rule engine**, mapped to Handbook clauses, decides compliance

Verify: a **Verification Ladder** runs issuer API → digital signature → QR → portal → manual review, and logs source, time and result

Report: an **Evidence Card** shows the highlighted proof, the clause and the next step

What makes it unique

Graph-based checks: catches contradictions between documents, not just errors inside one file

Rules decide, LLM only explains: results are repeatable and auditable

Forensic signals: file and perceptual hashing, PDF edit traces, signature validation, date-logic checks

Versioned and tamper-proof audit: every decision replays under its original rule version, with a hash-chained log

Maker-checker lock: no adverse action without officer sign-of

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TECHNICAL APPROACH

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1. Technologies Used

Tools & technologies powering the solution

Frontend
React / Next.js

Backend
Node.js / Express (REST APIs),
Python (FastAPI) for AI services

Database
PostgreSQL
(records, rules, audit log)

OCR + parsing
Tesseract / PaddleOCR +
layout parser

AI
Gemini API
(classify and explain only)

Anomaly checks
OpenCV (image hash, photo checks),
PyMuPDF (PDF metadata and edit traces)

Verification
Issuer APIs, QR/digital signature validation

Security
Role-based access (JWT),
AES encryption, access logs

Hardware
Standard cloud server,
no special hardware

2. Methodology (Flow Chart)

From document upload to verified evidence



3. Implementation Plan

Phased approach for successfiul deployment

Phase 1
OCR + extraction + checklist rules
(3 document types)
- PDFs
- Images
- Scans

Phase 2
Evidence Graph + date-logic checks
- Link records
- Validate dates
- Cross-reference

Phase 3
Verification Ladder + anomaly signals
- Multi-level checks
- Detect anomalies
- Risk scoring

Phase 4
Dashboard, alerts, audit log
- Real-time alerts
- Officer dashboard
- Full audit trail

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FEASIBILITY AND VIABILITY

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Feasibility

- **Technically ready:** OCR, hashing, PDF analysis and QR/signature checks already exist as open tools
- **Low cost:** runs on a standard cloud server, no special hardware
- **Low risk to start:** begins with 3 common document types and adds more later
- **Easy to plug in:** works beside the AICTE portal through REST APIs, no rebuild needed
- **Legally safe:** uses only authorized sources and role-based access

Challenge	Strategy
Issuer has no API	Fall back to QR or signature check, then manual review
Poor or blurry scans	Confidence score per field; low scores go to a human
False flags	Flags are “indicators” only; officer confirms
Data privacy	Role-based access, encryption, logs, retention limits
Rules change often	Editable rule configs with version history
Access to AICTE data	Demo on synthetic data; integration through authorized APIs only

Rollout : Prototype → Pilot on one application category → Scale to all categories

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IMPACT AND BENEFITS

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Impact on the Target Audience

Who	Impact
AICTE officers	No repeat checking; only exceptions reach them
Institutions	Instant “what’s missing” report; fix gaps before submission
Students and parents	Approvals rest on verified documents, so they can trust an institution’s status

Benefits

- **Social:** fair, evidence-based decisions. Good institutions are not wrongly flagged, and fake documents are caught
- **Economic:** less officer time per application, fewer returned applications and re-submissions, lower inspection cost since only flagged cases need visits
- **Environmental:** less paper and fewer physical document checks and travel
- **Governance:** every decision has an audit trail, so processing is transparent and consistent

How We Measure It

- Extraction accuracy
- Source verification success rate
- Time per verification
- Alert speed
- Drop in manual effort

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RESEARCH AND REFERENCES

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1. AICTE and Regulatory Sources

AICTE Approval Process Handbook (current edition): source for the checklists and rules. Download from https://www.aicte-india.org

AICTE Extension of Approval (EoA) guidelines: https://www.aicte-india.org

DigiLocker (issuer-verified documents): https://www.digilocker.gov.in

IT Act, 2000 (digital signatures) and DPDP Act, 2023 (data protection): https://www.meity.gov.in

2. Research Work

LayoutLM: Pre-training of Text and Layout for Document Image Understanding (Xu et al., 2020): https://arxiv.org/abs/1912.13318. This is the basis for layout-aware extraction.

Perceptual hashing for duplicate and near-duplicate image detection: https://phash.org

3. Tools and Documentation

Tesseract OCR: https://github.com/tesseract-ocr/tesseract

PaddleOCR: https://github.com/PaddlePaddle/PaddleOCR

OpenCV: https://opencv.org

PyMuPDF (PDF metadata and edit traces): https://pymupdf.readthedocs.io

Gemini API: https://ai.google.dev/gemini-api/docs

FastAPI: https://fastapi.tiangolo.com

PostgreSQL: https://www.postgresql.org